Lesson 63: Universal Approximation

AI/ML Mathematics

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1 Motivation and Overview

Suppose we wish to approximate a continuous function $f\colon [0,1]^d \to \mathbb{R}$—say a value function $V^*(s)$ over a compact state space, or a conditional density used in a generative model—using a neural network with a single hidden layer and m neurons. The network computes

$$x \mapsto \sum_{j=1}^{m} a_j\, \sigma(w_j^\top x + b_j),$$

where σ is a fixed activation function. A natural and fundamental question arises: *for which activations σ can such networks approximate any continuous target f to arbitrary accuracy, provided m is large enough?* If the answer is negative, the entire architecture is limited; if positive, the question shifts to efficiency (how large must m be?) and learnability (can we find the weights?). The tools developed in this lesson—Cybenko's theorem and Hornik's generalization—show that the answer is positive for essentially every activation used in practice, and they identify the precise mathematical conditions that make universal approximation possible.

Why Universal Approximation Matters for RL/ML

- **Value function approximation.** In deep RL, one approximates $V^*(s)$ or $Q^*(s,a)$ with a neural network. Universal approximation guarantees that the hypothesis class is rich enough that the approximation error can, in principle, be driven to zero.
- **Policy gradient methods.** Parameterizing policies $\pi_\theta(a \mid s)$ via neural networks requires that the network class be sufficiently expressive to represent near-optimal policies.
- **Generative models.** Variational autoencoders and normalizing flows rely on neural networks to approximate complex densities; universal approximation justifies using these architectures.
- **Architecture design.** Understanding *which* activations yield universal approximation guides the choice between sigmoid, ReLU, and other nonlinearities.

Prerequisites. Compact subsets of $\mathbb{R}^d$ and continuous functions on them (Lesson 2, Heine–Borel theorem). Basic properties of L^p spaces (Lesson 4). The Hahn–Banach theorem and its corollaries (Lesson 21, Theorem 3.1). The Riesz representation theorem for $C(K)^*$ (Lesson 20, Theorem 3.2). Familiarity with signed Borel measures and weak-$*$ convergence.

Learning objectives. After completing this lesson, you will be able to:

- **Define** single-hidden-layer networks and state precisely what "universal approximation" means in $C(K)$ and $L^p(\mu)$.
- **State and prove** Cybenko's theorem for sigmoidal activations using the Hahn–Banach theorem and the Riesz representation theorem.
- **State and prove** Hornik's generalization for non-polynomial activations via the discriminatory function argument.
- **Verify** that specific activations (ReLU, tanh, sigmoid, softplus) satisfy the conditions for universal approximation.
- **Explain** the limitations of universal approximation theorems: the curse of dimensionality, the absence of constructive weight-finding, and the gap between approximation and generalization.
- **Justify** why Barron's theorem (Lesson 64) is needed to obtain quantitative approximation rates.

2 References

- Bach [2024, Chapter 6] — Presents universal approximation in the context of learning theory; includes both Cybenko's and Hornik's results with modern notation.
- Cybenko [1989] — The original paper proving that sigmoidal networks are dense in $C(K)$.
- Hornik et al. [1989] — Extends Cybenko's result to general non-polynomial activations and L^p approximation.
- Pinkus [1999] — A comprehensive survey of approximation theory for multilayer perceptrons, including detailed proofs.
- Lu et al. [2017] — Modern results on the expressive power of bounded-width networks with ReLU activations.
- Hanin [2019] — Universal approximation for deep ReLU networks with bounded width.

3 Single Hidden Layer Networks

We begin by giving a precise definition of the function class produced by a single-hidden-layer feedforward network.

Definition 3.1 (Single-Hidden-Layer Network)**.** Let $\sigma\colon \mathbb{R} \to \mathbb{R}$ be a fixed measurable function (the *activation function*). For $d \geq 1$ and $m \geq 1$, define the class of *single-hidden-layer networks with m neurons* by

$$\mathcal{F}_m^\sigma \;=\; \left\{ x \mapsto \sum_{j=1}^{m} a_j\, \sigma(w_j^\top x + b_j) \;:\; a_j \in \mathbb{R},\; w_j \in \mathbb{R}^d,\; b_j \in \mathbb{R} \right\}.$$

The class of all such networks (with any number of neurons) is

$$\mathcal{F}^\sigma \;=\; \bigcup_{m=1}^{\infty} \mathcal{F}_m^\sigma.$$

Each function $f \in \mathcal{F}_m^\sigma$ has $m(d+2)$ parameters: m output weights a_j, m weight vectors $w_j \in \mathbb{R}^d$, and m biases b_j.

Remark 3.2. For each fixed m, the set $\mathcal{F}_m^\sigma$ is *not* a linear subspace of $C(K)$ (it is not closed under addition unless we increase m). However, $\mathcal{F}^\sigma$ is a linear subspace: if $f \in \mathcal{F}_m^\sigma$ and $g \in \mathcal{F}_n^\sigma$, then $f + g \in \mathcal{F}_{m+n}^\sigma \subseteq \mathcal{F}^\sigma$.

Definition 3.3 (Universal Approximation Property)**.** Let $K \subseteq \mathbb{R}^d$ be compact. We say the activation function σ has the *universal approximation property in $C(K)$* if $\mathcal{F}^\sigma$ is dense in $C(K)$ with respect to the supremum norm: for every $f \in C(K)$ and every $\varepsilon > 0$, there exist $m \geq 1$ and parameters $(a_j, w_j, b_j)_{j=1}^m$ such that

$$\sup_{x \in K} \left| f(x) - \sum_{j=1}^{m} a_j\, \sigma(w_j^\top x + b_j) \right| < \varepsilon.$$

Similarly, σ has the *universal approximation property in $L^p(\mu)$* (for a finite Borel measure μ on $\mathbb{R}^d$ and $1 \leq p < \infty$) if $\mathcal{F}^\sigma$ is dense in $L^p(\mu)$.

Example 3.4 (Common Activations)**.** The following are widely used activation functions:

(i) **Logistic sigmoid:** $\sigma(t) = \frac{1}{1+e^{-t}}$. Smooth, bounded, with $\sigma(t) \to 1$ as $t \to +\infty$ and $\sigma(t) \to 0$ as $t \to -\infty$.
(ii) **Hyperbolic tangent:** $\sigma(t) = \tanh(t) = \frac{e^t - e^{-t}}{e^t + e^{-t}}$. Smooth, bounded, $\sigma(t) \to \pm 1$ as $t \to \pm\infty$.
(iii) **ReLU:** $\sigma(t) = \max(0, t)$. Continuous but not bounded; piecewise linear.
(iv) **Softplus:** $\sigma(t) = \log(1 + e^t)$. Smooth approximation to ReLU; unbounded.

Example 3.5 (A Simple Network on $[0, 1]$)**.** Let $d = 1$, $K = [0, 1]$, $\sigma(t) = \frac{1}{1+e^{-t}}$ (logistic sigmoid), and $m = 2$. Consider

$$f(x) = a_1\, \sigma(w_1 x + b_1) + a_2\, \sigma(w_2 x + b_2).$$

With $w_1 = w_2 = 10$, $b_1 = -3$, $b_2 = -7$, $a_1 = 1$, $a_2 = -1$, the function f approximates a "bump" that is close to 1 on an interval around $x \approx 0.3$–0.7 and close to 0 elsewhere. This illustrates how two neurons can construct approximate indicator functions of intervals, a building block used in the proof of Cybenko's theorem.

Network Width and Value Function Quality

In deep RL, the number of neurons m in the hidden layer controls the expressiveness of the value function approximation. Universal approximation says that, for any continuous V^*, sufficiently large m yields an approximation within ε. However, the theorem says nothing about *how large* m must be—this quantitative question is addressed by Barron's theorem (Lesson 64) and by Rademacher complexity bounds (Lesson 60).

We now turn to the first major result: Cybenko's theorem, which resolves the universal approximation question for sigmoidal activations.

4 Cybenko's Theorem

Cybenko's 1989 result was the first rigorous proof that neural networks with sigmoidal activations are universal approximators. The proof strategy is indirect: one uses the Hahn–Banach theorem to reduce the density question to a property of signed measures, and then shows that the sigmoidal assumption forces any annihilating measure to be zero.

Definition 4.1 (Sigmoidal Function)**.** A measurable function $\sigma\colon \mathbb{R} \to \mathbb{R}$ is *sigmoidal* if

$$\lim_{t\to+\infty} \sigma(t) = 1 \qquad \text{and} \qquad \lim_{t\to-\infty} \sigma(t) = 0.$$

Example 4.2 (Sigmoidal Functions)**.** The logistic sigmoid $\sigma(t) = 1/(1 + e^{-t})$ is sigmoidal. The rescaled hyperbolic tangent $\sigma(t) = \frac{1}{2}(\tanh(t) + 1)$ is sigmoidal. The Heaviside step function $\sigma(t) = \mathbf{1}_{[0,\infty)}(t)$ is also sigmoidal (though discontinuous). Note that the standard tanh is *not* sigmoidal in the sense of Definition 4.1, since $\tanh(t) \to \pm 1$; however, it satisfies Hornik's weaker condition (Section 5).

The key concept in Cybenko's proof is the notion of a discriminatory function, which we now introduce.

Definition 4.3 (Discriminatory Function)**.** Let $K \subseteq \mathbb{R}^d$ be compact. A measurable function $\sigma\colon \mathbb{R} \to \mathbb{R}$ is called *discriminatory* (with respect to K) if the following holds: for any finite signed Borel measure μ on K,

$$\int_K \sigma(w^\top x + b)\, d\mu(x) = 0 \quad \text{for all } w \in \mathbb{R}^d,\ b \in \mathbb{R} \qquad \Longrightarrow \qquad \mu = 0.$$

In other words, σ is discriminatory if the only signed measure annihilated by all functions $x \mapsto \sigma(w^\top x + b)$ is the zero measure.

Lemma 4.4 (Sigmoidal Implies Discriminatory). *If $\sigma\colon \mathbb{R} \to \mathbb{R}$ is continuous and sigmoidal (Definition 4.1), then σ is discriminatory with respect to any compact $K \subseteq \mathbb{R}^d$.*

Proof. Let μ be a finite signed Borel measure on K such that

$$\int_K \sigma(w^\top x + b)\, d\mu(x) = 0 \qquad \text{for all } w \in \mathbb{R}^d,\ b \in \mathbb{R}. \tag{1}$$

We must show $\mu = 0$.

Step 1: Pointwise limit. Fix $w \in \mathbb{R}^d$ and $\theta \in \mathbb{R}$, and define the half-space $H_{w,\theta} = \{x \in \mathbb{R}^d : w^\top x + \theta > 0\}$. For $\lambda > 0$ and any $b \in \mathbb{R}$, consider

$$\sigma(\lambda(w^\top x + b)).$$

Since σ is sigmoidal, as $\lambda \to +\infty$:

$$\sigma(\lambda(w^\top x + b)) \ \to\ \begin{cases} 1 & \text{if } w^\top x + b > 0, \\ 0 & \text{if } w^\top x + b < 0, \\ \sigma(0) & \text{if } w^\top x + b = 0. \end{cases}$$

That is, $\sigma(\lambda(w^\top x + b)) \to \mathbf{1}_{H_{w,-b}}(x)$ for μ-almost every x (the hyperplane $\{x : w^\top x + b = 0\}$ has $|\mu|$-measure zero for all but countably many values of b).

Step 2: Passing to the limit. By (1), for every $\lambda > 0$ and $b \in \mathbb{R}$,

$$\int_K \sigma(\lambda(w^\top x + b))\, d\mu(x) = 0.$$

(Here we replace w by λw and use the fact that the condition holds for all w.) The function σ is bounded (since it is continuous and converges at $\pm\infty$), so by the bounded convergence theorem, taking $\lambda \to +\infty$:

$$\int_K \mathbf{1}_{H_{w,-b}}(x)\, d\mu(x) = 0,$$

i.e., $\mu(H_{w,-b}) = 0$ for all half-spaces $H_{w,-b}$ (for almost every b, and by a density argument for all b).

Step 3: Half-spaces generate the Borel σ-algebra. Since the open half-spaces $\{H_{w,\theta} : w \in \mathbb{R}^d, \theta \in \mathbb{R}\}$ generate the Borel σ-algebra $\mathcal{B}(\mathbb{R}^d)$, a signed measure that vanishes on all half-spaces must be the zero measure. To see this, note that finite intersections of half-spaces form all open convex polytopes, and these form a π-system generating $\mathcal{B}(\mathbb{R}^d)$. Since μ vanishes on this π-system, it vanishes on the generated σ-algebra by the π-λ theorem (Lesson 1, Theorem 3.3). Therefore $\mu = 0$. □

We now state and prove the main result.

Theorem 4.5 (Cybenko, 1989). *Let $K \subseteq \mathbb{R}^d$ be compact and let $\sigma\colon \mathbb{R} \to \mathbb{R}$ be any continuous discriminatory function. Then $\mathcal{F}^\sigma$ is dense in $C(K)$ with respect to the supremum norm $\|f\|_\infty = \sup_{x\in K} |f(x)|$.*

Proof. Suppose for contradiction that $\overline{\mathcal{F}^\sigma} \neq C(K)$, where the closure is in the supremum norm. Since $\mathcal{F}^\sigma$ is a linear subspace of $C(K)$ (Remark 3.2), its closure $\overline{\mathcal{F}^\sigma}$ is a closed linear subspace of $C(K)$ that is strictly contained in $C(K)$.

By the Hahn–Banach theorem (Lesson 21, Theorem 3.1), there exists a bounded linear functional $L \in C(K)^*$ such that $L \neq 0$ and $L(g) = 0$ for all $g \in \overline{\mathcal{F}^\sigma}$.

By the Riesz representation theorem for $C(K)$ (Lesson 20, Theorem 3.2), every bounded linear functional on $C(K)$ is represented by a finite signed Borel measure: there exists a finite signed Borel measure μ on K such that

$$L(g) = \int_K g(x)\, d\mu(x) \qquad \text{for all } g \in C(K).$$

In particular, for every $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$, the function $x \mapsto \sigma(w^\top x + b)$ belongs to $\mathcal{F}_1^\sigma \subseteq \mathcal{F}^\sigma$, so

$$\int_K \sigma(w^\top x + b)\, d\mu(x) = L(\sigma(w^\top \cdot + b)) = 0.$$

Since σ is discriminatory (Definition 4.3), this implies $\mu = 0$, hence $L = 0$. This contradicts $L \neq 0$.

Therefore $\overline{\mathcal{F}^\sigma} = C(K)$, i.e., $\mathcal{F}^\sigma$ is dense in $C(K)$. □

Cybenko's Universal Approximation Theorem

Combining Theorem 4.5 and Lemma 4.4: if σ is continuous and sigmoidal, then for any compact $K \subseteq \mathbb{R}^d$, any $f \in C(K)$, and any $\varepsilon > 0$, there exist $m \geq 1$ and parameters $(a_j, w_j, b_j)_{j=1}^m$ such that

$$\sup_{x \in K} \left| f(x) - \sum_{j=1}^m a_j\, \sigma(w_j^\top x + b_j) \right| < \varepsilon.$$

Example 4.6 (Approximating a Step Function)**.** Let $K = [0,1]$, $f(x) = \mathbf{1}_{[1/3,2/3]}(x)$ (which is not continuous, but can be approximated by continuous functions). Consider a continuous approximation f_δ with linear ramps of width δ at the endpoints. For $\varepsilon > 0$, Cybenko's theorem guarantees that f_δ can be approximated within ε by a network with sigmoidal activation. In practice, this is achieved by using neurons with large weights to create steep transitions, as in Example 3.5.

Example 4.7 (Constructive Illustration in 1D)**.** To approximate $f(x) = x^2$ on $[0,1]$ using the logistic sigmoid $\sigma(t) = 1/(1+e^{-t})$, one can partition $[0,1]$ into m equal intervals and use $2m$ neurons to create approximate bump functions on each interval, scaling each bump by the value of f at the interval midpoint. This piecewise-constant approximation achieves $\|f - f_m\|_\infty \leq \frac{1}{4m^2}$ (since $f'' = 2$ gives error $\frac{1}{2} \cdot \frac{f''}{8} \cdot (1/m)^2$). The number of neurons grows as $O(1/\varepsilon^{1/2})$ for this particular smooth target.

Remark 4.8. Cybenko's proof is *non-constructive*: it shows that a good approximation exists but provides no algorithm for finding the weights (a_j, w_j, b_j). The Hahn–Banach theorem yields existence via contradiction, not via an explicit construction. This limitation motivates both (a) constructive proofs based on sampling (Barron's theorem, Lesson 64) and (b) the study of optimization algorithms for training (Lesson 65).

Having established Cybenko's theorem for sigmoidal activations, we next ask: what happens for activations that are not sigmoidal?

5 Hornik's Generalization

Cybenko's theorem covers sigmoidal activations (those converging to 0 and 1 at $\pm\infty$), but the most popular activation in modern deep learning—the ReLU $\sigma(t) = \max(0,t)$—is not sigmoidal. Hornik, Stinchcombe, and White (1989) showed that universal approximation holds for a much broader class of activations.

Definition 5.1 (Non-Polynomial Activation)**.** A continuous function $\sigma\colon \mathbb{R} \to \mathbb{R}$ is called *non-polynomial* if σ is not a polynomial, i.e., there is no polynomial p such that $\sigma(t) = p(t)$ for all $t \in \mathbb{R}$.

Remark 5.2. If σ is a polynomial of degree k, then every function in $\mathcal{F}^\sigma$ is also a polynomial of degree at most k in each coordinate. In particular, $\mathcal{F}^\sigma$ cannot approximate all continuous functions. Therefore the non-polynomial condition is *necessary* for universal approximation.

We now show that continuous non-polynomial activations are discriminatory, which by Cybenko's framework (Theorem 4.5) immediately implies universal approximation.

Lemma 5.3 (Non-Polynomial Implies Discriminatory)**.** *If $\sigma\colon \mathbb{R} \to \mathbb{R}$ is continuous and non-polynomial, then σ is discriminatory with respect to any compact $K \subseteq \mathbb{R}^d$.*

Proof. Let μ be a finite signed Borel measure on K satisfying

$$\int_K \sigma(w^\top x + b)\, d\mu(x) = 0 \qquad \text{for all } w \in \mathbb{R}^d,\ b \in \mathbb{R}. \tag{2}$$

We show that $\mu = 0$.

Step 1: The annihilating condition extends to polynomials. Define the function $F\colon \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}$ by

$$F(w, b) = \int_K \sigma(w^\top x + b)\, d\mu(x).$$

By (2), $F(w, b) = 0$ for all (w, b). Since σ is continuous and K is compact, F is infinitely differentiable in (w, b), and we can differentiate under the integral sign. In particular, for any multi-index $\alpha = (\alpha_1, \ldots, \alpha_d)$ and any $n \geq 0$,

$$\frac{\partial^{|\alpha|+n} F}{\partial w_1^{\alpha_1} \cdots \partial w_d^{\alpha_d}\, \partial b^n}(w, b) = \int_K x_1^{\alpha_1} \cdots x_d^{\alpha_d}\, \sigma^{(|\alpha|+n)}(w^\top x + b)\, d\mu(x) = 0,$$

where $\sigma^{(k)}$ denotes the k-th derivative of σ.

Step 2: Evaluating at the origin. Setting $w = 0$ and $b = 0$:

$$\sigma^{(|\alpha|+n)}(0) \int_K x_1^{\alpha_1} \cdots x_d^{\alpha_d}\, d\mu(x) = 0.$$

Since σ is not a polynomial, its derivatives of all orders cannot all vanish at a single point: there exists a sequence of orders k_j such that $\sigma^{(k_j)}(0) \neq 0$. More precisely, if $\sigma^{(k)}(0) = 0$ for all $k \geq N$ for some N, then σ would be a polynomial of degree at most $N - 1$ in a neighborhood of 0 (by Taylor's theorem), and by analyticity arguments on each interval, σ would be a polynomial globally—contradicting the assumption.

Therefore, for each choice of $|\alpha| + n$, we can find a derivative order where σ does not vanish. Specifically, for any monomial exponent $(p_1, \ldots, p_d)$ with $p_1 + \cdots + p_d = k$, we choose $\alpha = (p_1, \ldots, p_d)$ and $n = 0$ and pick $w = 0$, $b = 0$; if $\sigma^{(k)}(0) = 0$, we instead choose $n > 0$ so that $|\alpha| + n = k + n$ where $\sigma^{(k+n)}(0) \neq 0$, and absorb the extra b-derivatives. After dividing by the nonzero constant $\sigma^{(k+n)}(0)$, we obtain

$$\int_K x_1^{p_1} \cdots x_d^{p_d}\, d\mu(x) = 0 \qquad \text{for all } p_1, \ldots, p_d \geq 0. \tag{3}$$

Step 3: Moment characterization. Equation (3) says that all moments of μ vanish. Since K is compact, μ is a compactly supported signed measure, and the Weierstrass approximation theorem (or Stone–Weierstrass) implies that polynomials are dense in $C(K)$. Therefore, for any $g \in C(K)$, we can approximate g uniformly by polynomials p_n, and

$$\int_K g(x)\, d\mu(x) = \lim_{n\to\infty} \int_K p_n(x)\, d\mu(x) = 0.$$

Since this holds for all $g \in C(K)$, the Riesz representation theorem implies $\mu = 0$. □

Combining Lemma 5.3 with Theorem 4.5, we obtain:

Theorem 5.4 (Hornik, Stinchcombe, and White, 1989)**.** *Let $K \subseteq \mathbb{R}^d$ be compact and let $\sigma\colon \mathbb{R} \to \mathbb{R}$ be any continuous, non-polynomial function. Then $\mathcal{F}^\sigma$ is dense in $C(K)$ with respect to the supremum norm. More generally, $\mathcal{F}^\sigma$ is dense in $L^p(\mu)$ for any finite Borel measure μ on K and $1 \le p < \infty$.*

Proof. The $C(K)$ density follows directly from Theorem 4.5 and Lemma 5.3: since continuous non-polynomial activations are discriminatory, Cybenko's framework applies.

For the $L^p(\mu)$ extension: since $C(K)$ is dense in $L^p(\mu)$ (as K is compact), and $\mathcal{F}^\sigma$ is dense in $C(K)$, transitivity of density gives that $\mathcal{F}^\sigma$ is dense in $L^p(\mu)$. □

Hornik's Universal Approximation Theorem

For any continuous non-polynomial activation σ (including ReLU, tanh, logistic sigmoid, softplus, ELU, swish, and essentially every activation used in practice), single-hidden-layer networks are dense in $C(K)$ for compact K and in $L^p(\mu)$ for any finite measure μ. The only activations that fail are polynomials.

Example 5.5 (ReLU Is a Universal Approximator)**.** The ReLU activation $\sigma(t) = \max(0, t)$ is continuous and non-polynomial (it is not differentiable at $t = 0$, so it cannot be a polynomial). Therefore, by Theorem 5.4, ReLU networks are universal approximators.

Concretely, a ReLU network $f_m(x) = \sum_{j=1}^m a_j \max(0, w_j^\top x + b_j)$ computes a piecewise-affine function on $\mathbb{R}^d$. The set of "breakpoints" (where some neuron transitions between its active and inactive regimes) divides $\mathbb{R}^d$ into convex polytopes, and f_m is affine on each polytope. As $m \to \infty$, the number of polytopes grows and the piecewise-affine function can approximate any continuous function.

Example 5.6 (Verification for tanh)**.** The hyperbolic tangent $\sigma(t) = \tanh(t)$ is continuous and non-polynomial (its Taylor series $\tanh(t) = t - t^3/3 + 2t^5/15 - \cdots$ has infinitely many nonzero terms and converges on all of $\mathbb{R}$; a polynomial would have only finitely many). Therefore tanh is a universal approximator by Theorem 5.4.

Density Does Not Imply Efficient Approximation

Universal approximation guarantees that for every $\varepsilon > 0$ there *exists* a network with finitely many neurons achieving error less than ε. It does *not* bound the number of neurons m as a function of ε and d. For generic continuous functions on $[0, 1]^d$, achieving ε-accuracy may require $m = \Omega(\varepsilon^{-d})$ neurons—an exponential dependence on dimension. This is the *curse of dimensionality*, which motivates Barron's theorem (Lesson 64).

We now examine precisely what universal approximation theorems do and do not tell us.

6 Limitations of Universal Approximation

Universal approximation is an existence theorem: it guarantees that neural networks form a rich enough function class. However, three critical gaps remain between the theoretical guarantee and practical neural network performance.

6.1 Gap 1: Approximation Rate (The Curse of Dimensionality)

Proposition 6.1 (Lower Bound for Generic Functions)**.** *Let $\mathcal{F}_m^\sigma$ denote the class of single-hidden-layer networks with m neurons and any continuous non-polynomial activation. For the*

class of all L-Lipschitz functions on $[0,1]^d$, the minimax approximation error satisfies

$$\inf_{f_m \in \mathcal{F}_m^\sigma} \sup_{\substack{f:[0,1]^d \to \mathbb{R} \\ L\text{-}Lipschitz}} \|f - f_m\|_{L^2} \;\geq\; c\,L\,m^{-1/d}$$

for a constant $c > 0$ depending only on d. That is, approximating generic Lipschitz functions to accuracy ε requires $m = \Omega(\varepsilon^{-d})$ neurons.

Proof. This follows from classical lower bounds in approximation theory. The class of L-Lipschitz functions on $[0,1]^d$ has ε-covering number (in L^2) at least $\exp(c'(L/\varepsilon)^d)$ for a constant $c' > 0$. On the other hand, $\mathcal{F}_m^\sigma$ has $m(d+2)$ parameters, so its covering number is at most $\exp(c''\,m(d+2)\log(1/\varepsilon))$. Matching these gives $m \geq c\,(L/\varepsilon)^d/((d+2)\log(1/\varepsilon))$, which yields the stated $m^{-1/d}$ rate. □

Example 6.2 (Dimension Scaling)**.** To approximate a Lipschitz function on $[0,1]^{100}$ to accuracy $\varepsilon = 0.01$ in the worst case, the lower bound requires $m = \Omega(100^{100})$ neurons—an astronomically large network. In contrast, Barron's theorem (Lesson 64) shows that for functions in the Barron class, $m = O(1/\varepsilon^2)$ suffices regardless of d, dramatically escaping this curse.

6.2 Gap 2: Finding the Weights (Optimization)

Even when a good approximation exists, the training problem

$$\min_{a_j, w_j, b_j} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{m} a_j\, \sigma(w_j^\top x_i + b_j) \right)^2$$

is non-convex in the parameters (w_j, b_j) whenever σ is nonlinear. Universal approximation says nothing about whether gradient descent or any other algorithm can find a (near-)global minimum.

Remark 6.3. Recent work on the *neural tangent kernel* (NTK) [Jacot et al., 2018] shows that, in the infinite-width limit ($m \to \infty$), gradient descent on overparameterized networks converges to a global minimum of the training loss. This connects the approximation theory of this lesson to the optimization theory of Lesson 65. However, the NTK regime requires m to be very large (polynomial in n), and it is debated whether this regime captures the practical behavior of finite-width networks.

6.3 Gap 3: Generalization from Finite Data

Approximation error measures how well the best network in $\mathcal{F}_m^\sigma$ can represent the target function. But in practice, we only have n training samples and must also control the *estimation error*: the gap between the best-in-class network and the network we actually find from data. The total error decomposes as

$$\underbrace{R(\hat{f}_n) - \inf_{f \in \mathcal{F}} R(f)}_{\text{estimation error}} \;+\; \underbrace{\inf_{f \in \mathcal{F}} R(f) - \inf_{f} R(f)}_{\text{approximation error}},$$

where $R(f) = \mathbb{E}[\ell(f(X), Y)]$ is the population risk. Universal approximation controls the second term, but the first term depends on the complexity of $\mathcal{F}_m^\sigma$ (measured, e.g., by Rademacher complexity). Making m larger reduces approximation error but increases estimation error—a fundamental tradeoff analyzed in Lesson 60.

The Bias-Variance Tradeoff in RL

In deep RL, the same tradeoff appears:

- A very large network ($m \gg 1$) can represent V^* well (*low approximation error*), but may overfit to sampled transitions (*high estimation error*).
- A small network may underfit V^* (*high approximation error*) but generalize well from limited data.

Architecture selection in DQN, PPO, and SAC implicitly balances these competing effects.

Example 6.4 (Overfitting with Universal Approximators)**.** A network with $m = n$ neurons can interpolate n arbitrary data points (x_i, y_i) exactly (under mild conditions on the x_i). This means the training error is zero, yet the test error can be arbitrarily large. Universal approximation guarantees richness, not generalization.

The limitations identified in this section—exponential width dependence on dimension, non-convex optimization, and the approximation-estimation tradeoff—motivate the study of Barron's theorem in Lesson 64, which provides dimension-free approximation rates for a natural subclass of functions.

7 Exercises

Exercise 7.1 (Discriminatory Verification)**.** (*)
Show that the Heaviside step function $\sigma(t) = \mathbf{1}_{[0,\infty)}(t)$ is discriminatory, directly from Definition 4.3, without using Lemma 4.4.
Hint: If $\int_K \mathbf{1}_{[0,\infty)}(w^\top x + b)\, d\mu(x) = 0$ for all w, b, show that $\mu(H) = 0$ for every open half-space H.

Exercise 7.2 (Polynomial Activations Fail)**.** (*)
Let $\sigma(t) = t^2$. Show that $\mathcal{F}^\sigma$ is *not* dense in $C([0,1])$. Specifically, find a continuous function on $[0,1]$ that cannot be approximated by functions of the form $\sum_{j=1}^m a_j(w_j x + b_j)^2$, and prove your claim.

Exercise 7.3 (ReLU Piecewise Affine Representation)**.** (**)
Let $d = 1$ and $\sigma(t) = \max(0, t)$ (ReLU).

(a) Show that any $f \in \mathcal{F}_m^\sigma$ is a continuous piecewise-affine function on $\mathbb{R}$ with at most m breakpoints.

(b) Conversely, show that any continuous piecewise-affine function on $\mathbb{R}$ with k breakpoints belongs to $\mathcal{F}_{k+1}^\sigma$.

(c) Using part (b), prove directly (without Theorem 5.4) that ReLU networks are dense in $C([a,b])$ for any $[a,b]$.

Exercise 7.4 (Softplus Approximation of ReLU)**.** (**)
The softplus activation is $\sigma_\beta(t) = \frac{1}{\beta}\log(1 + e^{\beta t})$ for $\beta > 0$.

(a) Show that $\sigma_\beta(t) \to \max(0, t)$ pointwise as $\beta \to \infty$.

(b) Show that the convergence is uniform on compact subsets of $\mathbb{R} \setminus \{0\}$ but *not* uniform on any neighborhood of 0.

(c) Conclude that if softplus networks are universal approximators (which they are by Theorem 5.4), then ReLU networks are also universal approximators, giving an alternative proof for ReLU.

Exercise 7.5 (Value Function Approximation in RL). (**)
Consider a continuous MDP with state space $K = [0,1]^d$ and a continuous optimal value function $V^*\colon K \to [0, V_{\max}]$. Let $\hat{V}_m$ be the best approximation of V^* in $\mathcal{F}_m^\sigma$ (with $\sigma = \text{ReLU}$) in the supremum norm.

(a) State precisely what Theorem 5.4 guarantees about $\|V^* - \hat{V}_m\|_\infty$ as $m \to \infty$.

(b) Explain why this guarantee alone is insufficient for practical deep RL. Identify at least two additional quantities that must be controlled.

(c) If V^* belongs to the Barron class (defined in Lesson 64), state the improved approximation rate.

Exercise 7.6 (Universal Approximation in L^2 but not C). (***)
Construct a bounded measurable (but discontinuous) activation σ that is discriminatory but for which $\mathcal{F}^\sigma$ is *not* contained in $C(K)$ (and therefore the $C(K)$-density statement does not apply, though L^2-density may still hold). Specifically, let $\sigma = \mathbf{1}_{[0,\infty)}$ and show:

(a) Every $f \in \mathcal{F}^\sigma$ is a step function (finite combination of indicator functions of half-spaces).

(b) $\mathcal{F}^\sigma$ is dense in $L^2(\mu)$ for any compactly supported absolutely continuous measure μ.

(c) $\mathcal{F}^\sigma$ is *not* dense in $C(K)$ because elements of $\mathcal{F}^\sigma$ are not continuous.

8 Summary

- A single-hidden-layer network with m neurons computes $f(x) = \sum_{j=1}^m a_j \sigma(w_j^\top x + b_j)$; the class $\mathcal{F}^\sigma = \bigcup_m \mathcal{F}_m^\sigma$ forms a linear subspace of the function space.
- An activation σ is *discriminatory* if the only signed measure annihilated by all functions $x \mapsto \sigma(w^\top x + b)$ is the zero measure.
- **Cybenko's theorem:** If σ is continuous and discriminatory (which holds for all continuous sigmoidal functions), then $\mathcal{F}^\sigma$ is dense in $C(K)$ for compact K. The proof uses the Hahn–Banach theorem and the Riesz representation theorem.
- **Hornik's theorem:** The discriminatory condition holds for all continuous non-polynomial activations. Therefore, any continuous non-polynomial σ (including ReLU, tanh, sigmoid, softplus) yields universal approximation in $C(K)$ and $L^p(\mu)$.
- The only activations that fail to be universal approximators are polynomials, for which the output is necessarily polynomial.
- Universal approximation is a *qualitative* result: it guarantees density but does not specify (a) how many neurons are needed, (b) how to find the weights, or (c) how the network generalizes from finite data.
- For generic Lipschitz functions on $[0,1]^d$, achieving ε-accuracy requires $m = \Omega(\varepsilon^{-d})$ neurons in the worst case—an exponential dependence on dimension.
- Barron's theorem (Lesson 64) escapes this curse for a restricted but natural class of functions, providing a dimension-free $O(1/m)$ rate.

9 Further Reading

- **Lesson 64** extends the analysis to quantitative approximation rates via Barron's theorem, showing that for functions with bounded Fourier moment, the rate $O(1/m)$ is dimension-free.
- **Lesson 65** addresses the optimization question: how gradient-based methods find good parameters in practice, including the role of overparameterization and the neural tangent kernel.
- Pinkus [1999] provides a comprehensive survey of approximation-theoretic results for neural networks, including generalizations to multiple hidden layers.
- Telgarsky [2016] and Eldan and Shamir [2016] study the *depth* advantage: there exist functions expressible by depth-k networks with polynomial width that require exponential width at depth $k-1$. This motivates deep (multi-layer) architectures beyond the single-layer setting of this lesson.
- Hanin [2019] proves universal approximation for deep ReLU networks with *bounded width* (width $d+1$ suffices for L^p approximation on $\mathbb{R}^d$), complementing the width-based results studied here.
- **Historical note.** Cybenko's 1989 paper was motivated by the question of whether "connectionist" models (the terminology of the era) had fundamental representational limitations. Hornik, Stinchcombe, and White's concurrent 1989 paper gave a more general answer. Both built on classical functional analysis—particularly the Hahn–Banach theorem—showing a beautiful application of pure mathematics to a question from machine learning.

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